Evaluating Traditional and Large Language Model Approaches for Aspect-Based Sentiment Analysis of Business Reviews

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WI26 Data Science Capstone
2nd Progress Report
GitHub

Section 1: Project Summary

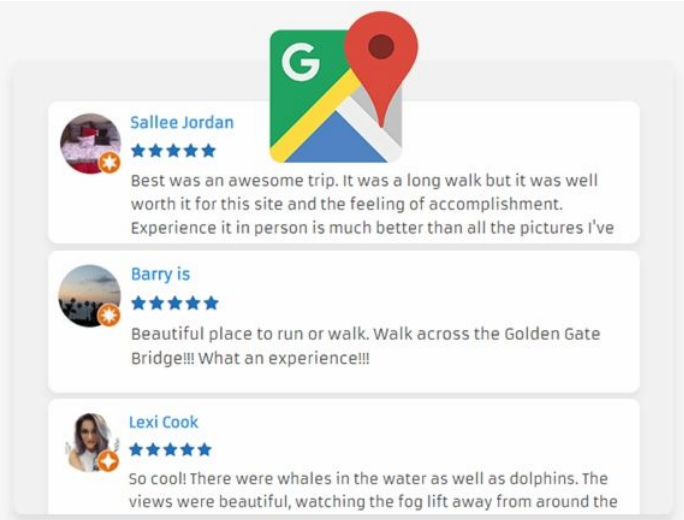
Background

The Challenge

- Platforms like Google Maps generate **millions** of customer reviews
- Rich, but **unstructured** feedback
- Manual analysis is slow, inconsistent, & not scalable

Why It Matters

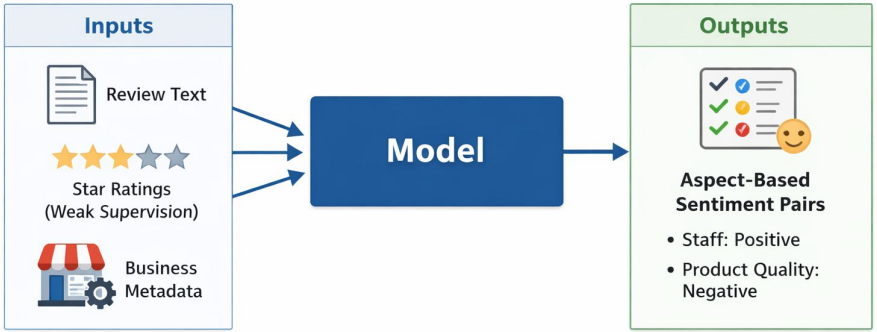
- Reviews impact reputation & revenue
- Valuable insights buried in large text volumes
- Businesses need scalable, consistent solutions for enabling data-driven business decisions



Problem Definition

- Analyze large-scale business review text to extract aspect-based sentiment analysis (ABSA) insights that identify top strengths and improvement areas for businesses.

- Develop and evaluate ABSA pipelines using:
  1. Feature-engineered traditional natural language processing (NLP) models
  2. Neural transformer-encoder architectures, and
  3. Generative large language models (LLMs)



Why ML?
This problem involves massive, linguistically complex text data that exceeds manual & rule-based analysis capabilities.

Dataset: Google Local Reviews (2021)

Source: Zhang, T., & Li, J. (2021). *Google Local Data (2021)*[Data set]. UC San Diego McAuley Lab. Retrieved from https://mcauleylab.ucsd.edu/public_datasets/gdrive/googlelocal/

Dataset consists of two JSON files per U.S. State:

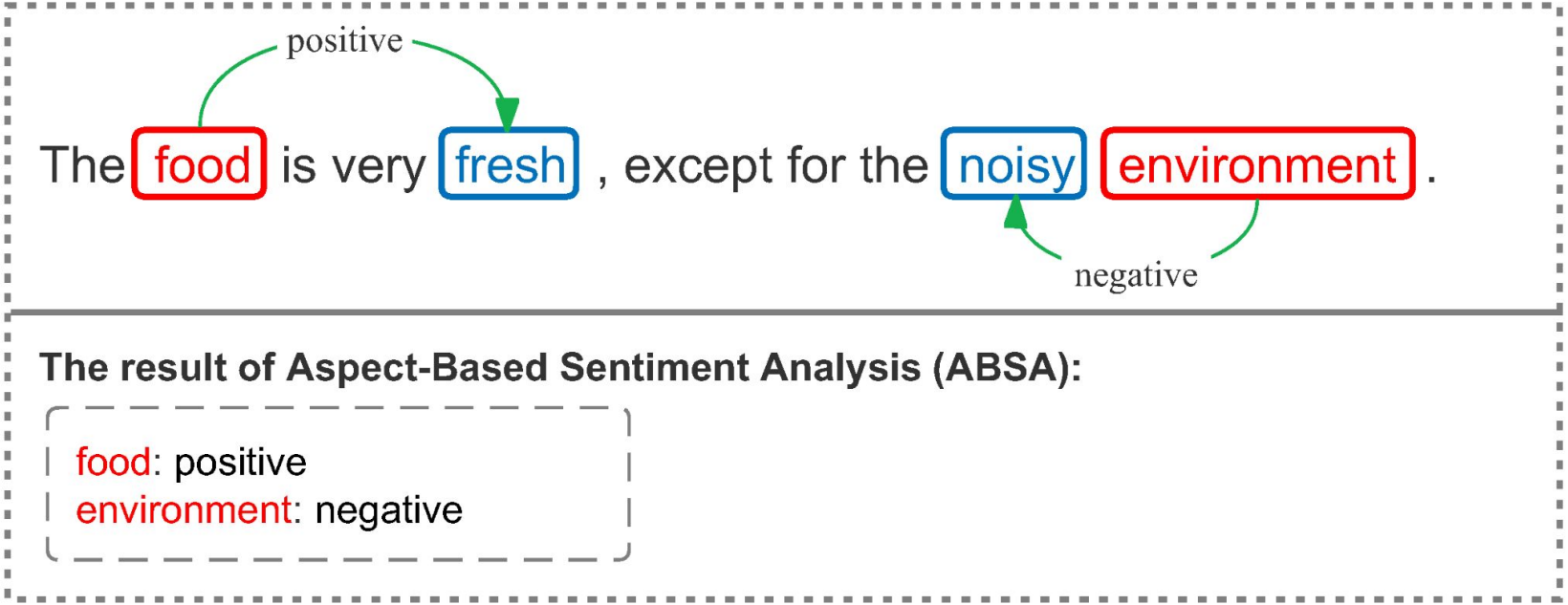
- **reviews.json**: user_id, rating (1-5), text (review content), gmap_id (business id), timestamp
- **metadata.json**: gmap_id, business name, description, additional business attributes

Full dataset = 650+ million reviews for all 50 U.S. States (200GB)

Our subset = California only, 70+ million reviews for 500,000+ businesses

	user_id	time	rating	text	gmap_id	name_business	category
0	1.131656e+20	1599164133778	5.0	Great company. Amazing customer service and th...	0x80c2c98c0e3c16fd:0x29ec8a728764fdf9	City Textile	['Textile exporter']
1	1.012264e+20	1618261672851	5.0	Nice people\nhelpful	0x80c2c98c0e3c16fd:0x29ec8a728764fdf9	City Textile	['Textile exporter']

Aspect-Based Sentiment Analysis (ABSA)

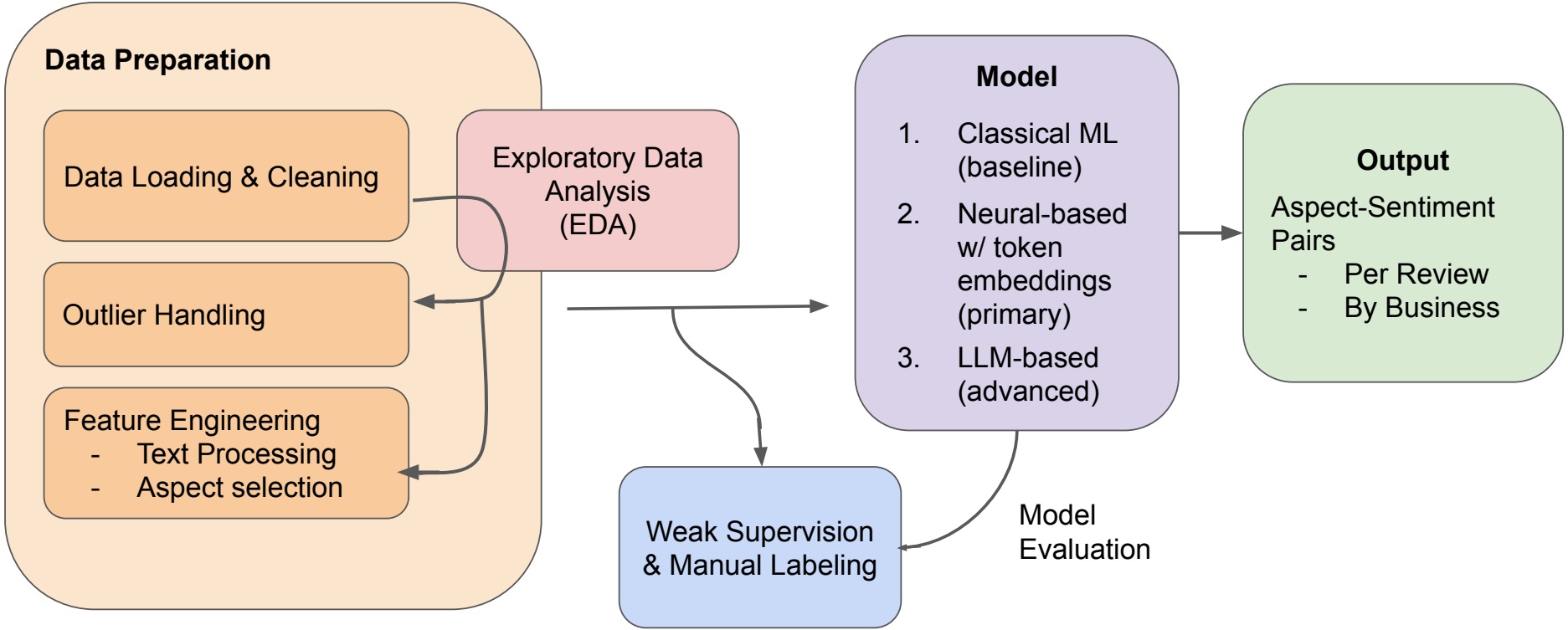


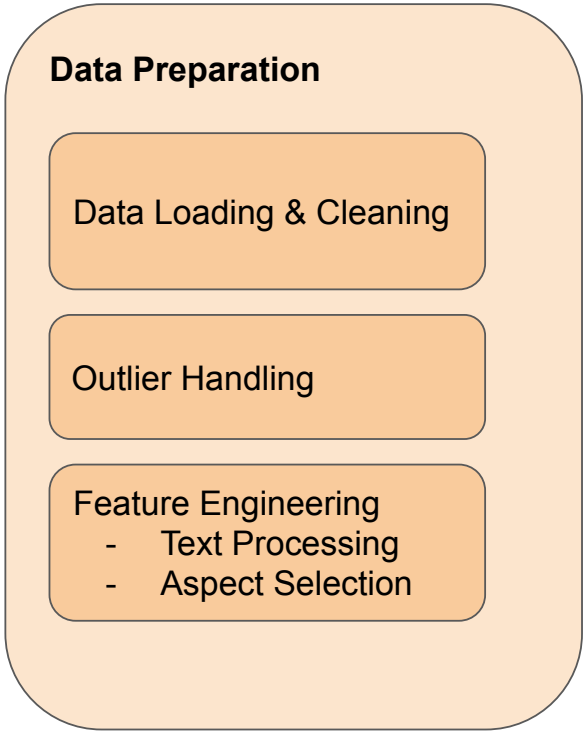
Source: Xue, J., Li, Y., Li, Z., Cui, Y., Zhang, S., & Wang, S. (2023). *A Cross-Domain Generative Data Augmentation Framework for Aspect-Based Sentiment Analysis*. Electronics, 12(13), 2949. https://doi.org/10.3390/electronics12132949

Literature Survey: ABSA

- 📖 Rule-based Methods (Early 2000s)[1]
  - Extract aspects via noun frequency
  - Assign sentiment using polarity lexicons
- 📉Traditional Machine Learning (2010s)[2, 5]
  - Supervised & semi-supervised models (SVM, logistic regression)
  - Learned aspect-sentiment patterns from labelled data
  - Requires manually labelled data, time-intensive
- 🧠Deep Learning & LLMs (2020s)[3, 4, 5]
  - Neural architectures (transformers)
  - Pretrained language models for ABSA
  - High computational cost and complexity

Google Reviews ABSA Pipeline

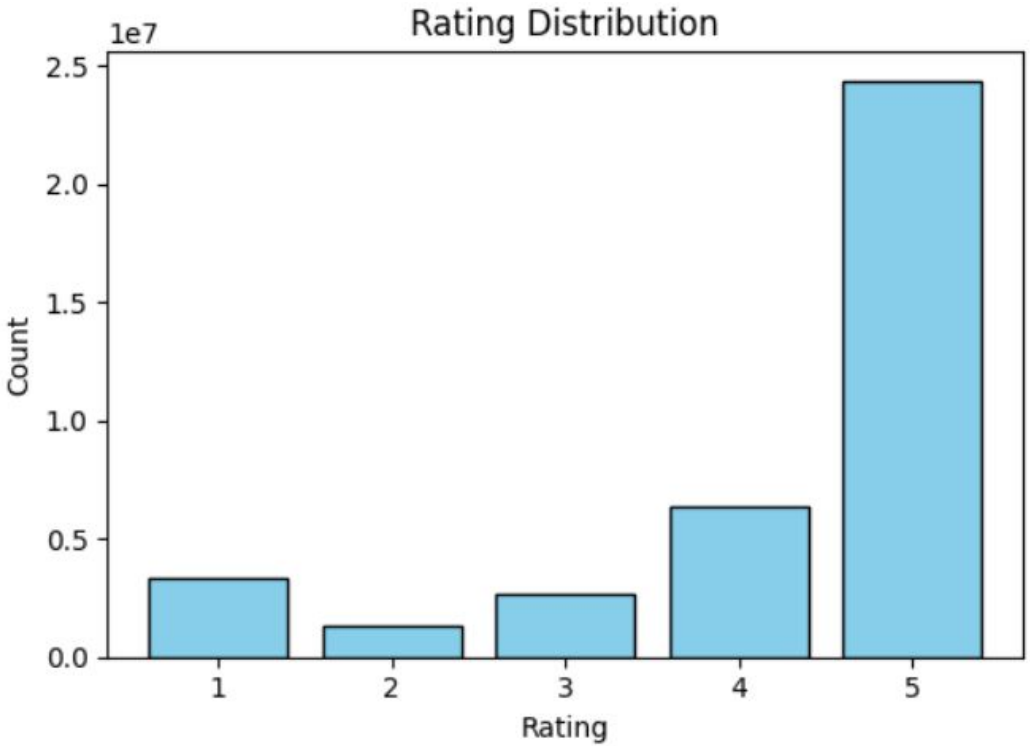


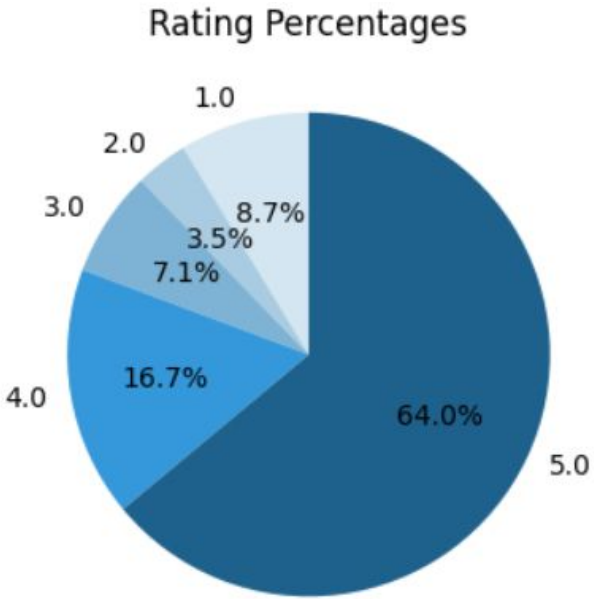


- Data Loading & Cleaning
  - Load Raw Data (JSON): 70,632,313 reviews for 515,961 businesses
  - Filter columns ['user_id', 'rating', 'text', 'gmap_id', 'time']
  - Remove duplicates, missing ratings, empty review text, non-English reviews
- Outlier Handling (Post-EDA)
  - Filter for users with 5-250 reviews (remove sparse and spam users)
  - Filter for businesses with 10+ reviews (remove unreliable businesses)
  - Filter for reviews from 2011+ (10-year recency, pre-2010 is 0.02% of data)
- Feature Engineering
  - Add unique review_id
  - Add model-specific text processing
    - Lowercasing
    - Remove punctuation
    - Sentence tokenization
    - Transformer encoding
  - Add features for weak supervision
    - Aspect extraction (keywords in review text)
    - Sentiment extraction (ratings)

Exploratory Data Analysis (EDA) Summary

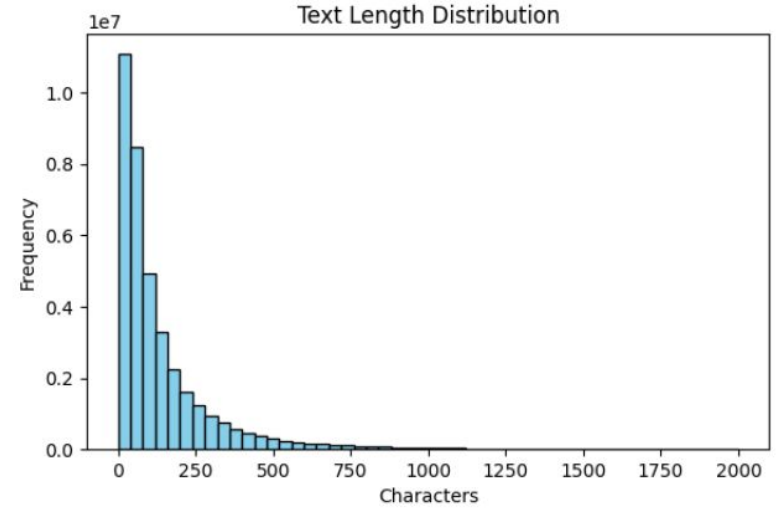
Exploratory Data Analysis (EDA)

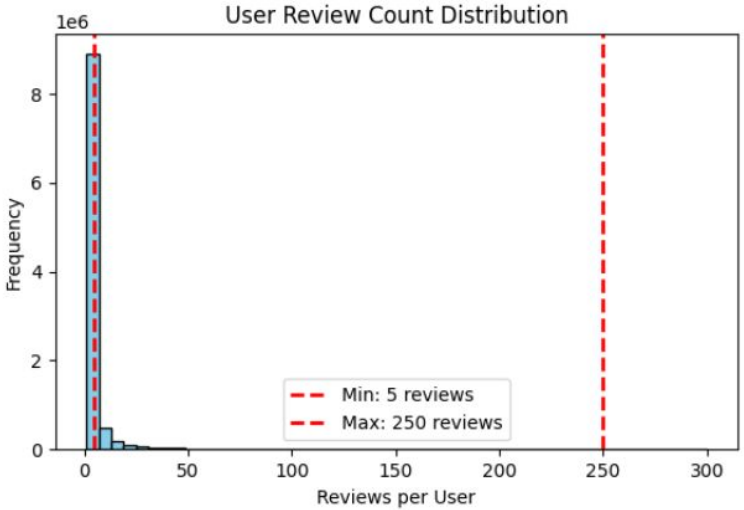


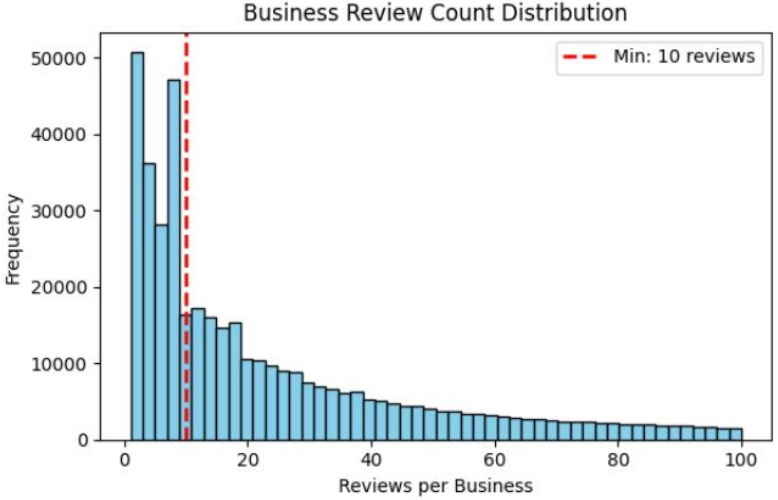


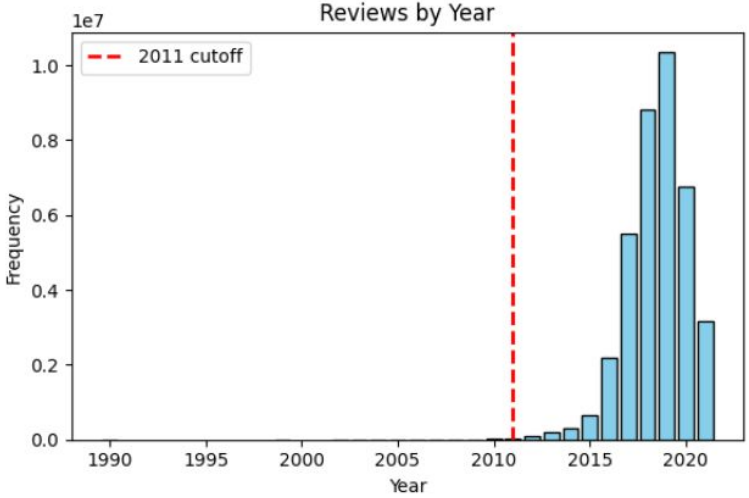
Exploratory Data Analysis (EDA)

Outlier Handling









Exploratory Data Analysis (EDA)

Text-based EDA Summary

Word Frequency by Rating

Negative reviews (1-2 stars): Narrative complaint words - `told`, `said`, `asked`, `never`, `didn`, Aspect words `service` and `order` , Sentiment words are strongly negative: `bad`, `rude`, `worst`, `horrible`.

Positive reviews (4-5 stars): Strong sentiment words - `nice`, `friendly`, `clean`, `fast`, `best`, `amazing`, `excellent`, `awesome`, `delicious`, `helpful`.

Key Takeaways for Aspect Modeling

1. **Aspect keywords are rating-agnostic** - `food`, `service`, `place`, `staff` appear across all ratings.
2. **Sentiment keywords are rating-specific** - `rude`, `worst`, `horrible` skew negative; `amazing`, `excellent`, `delicious` skew positive.
3. **Service is the most discriminative aspect** - it appears prominently across all ratings but with very different associated words (negative: `rude`, `customer service`; positive: `friendly`, `helpful`)
4. **Dataset scope affects aspect design** - with 86,882 unique category combinations spanning restaurants, retail, hotels, parks and more, aspect keywords need to be general enough to apply across business types

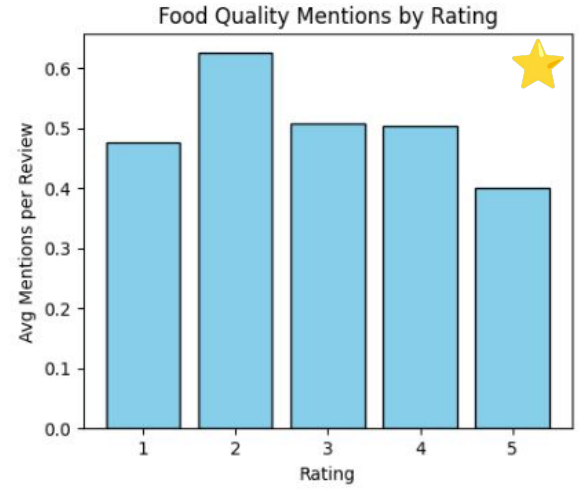
Feature Engineering
- Text Processing
- Aspect Selection

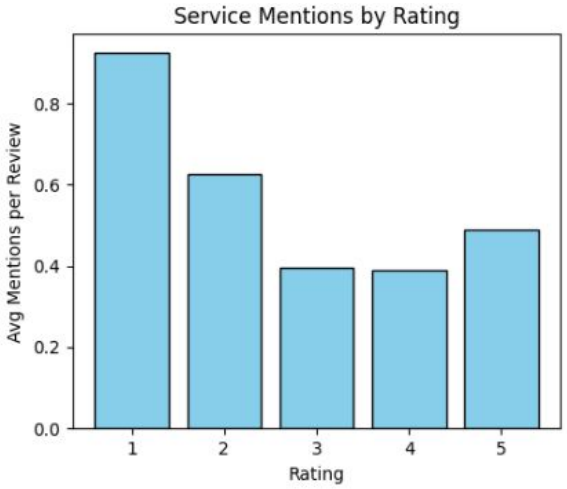
Aspect-Selection

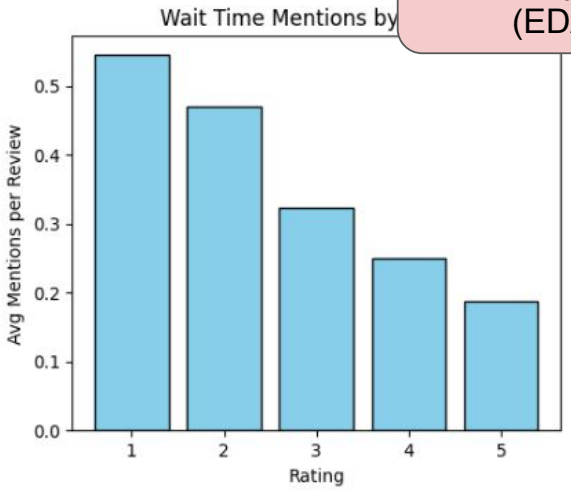
Used word-frequency EDA to select the following aspects for the dataset:

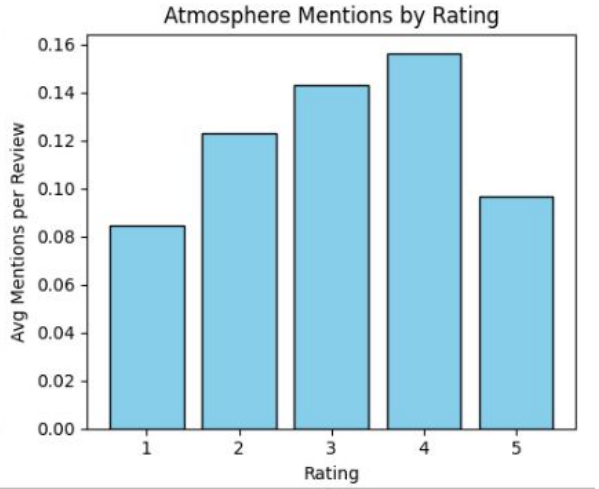
- Product Quality ⭐
- Service 🤝
- Wait Time 🕐
- Price/Value 💲
- Cleanliness 🧼
- Atmosphere 🎵
- General 💬

Exploratory Data Analysis (EDA)









Weak Supervision & Manual Labeling

Labeling (Two Variations)

1. Weak Supervision (allows for quick label assignments on all data)
   a. **Aspects** extracted via key word list (e.g., 'price_value': ['price', 'prices', 'expensive', 'cheap', 'cost', 'value', 'worth', 'affordable', 'overpriced', 'reasonable', 'pricey', 'money', 'deal']
   b. **Sentiment** is determined by the given rating (1-3 = negative, 4-5 = positive)
2. Manual Labelling (on subset of 10,000 reviews, balanced rating class)
   a. **Aspect-sentiment pairs** assigned to sentence tokens by a **human reviewer**
   b. Developed a widget to help with fast labeling

Model 1: Classical ML ABSA (baseline)

- Term Frequency - Inverse Document Frequency (TF-IDF) (sklearn)
- 90/10 train-test split (weakly supervised data)

$$TF(t,d) = \frac{\#\ of\ occurrences\ of\ term\ \ t\ in\ document\ d}{total\ \#\ of\ terms\ in\ document\ d}$$

$$IDF(t,D) = log\frac{Total\ \#\ of\ documents\ in\ the\ corpus}{\#\ of\ documents\ with\ the\ term\ t\ in\ them}$$

$$TFIDF(t,d,D) = TF(t,d)\ x\ IDF(t,D)$$

```
tfidf = TfidfVectorizer(
    max_features=10000,
    stop_words=stop_words,
    min_df=500,
    max_df=0.5,
    ngram_range=(1, 2)
    )
```

```
stop_words = [w for w in text.ENGLISH_STOP_WORDS if w not in ('not', 'no', 'never', 'nor')]
```

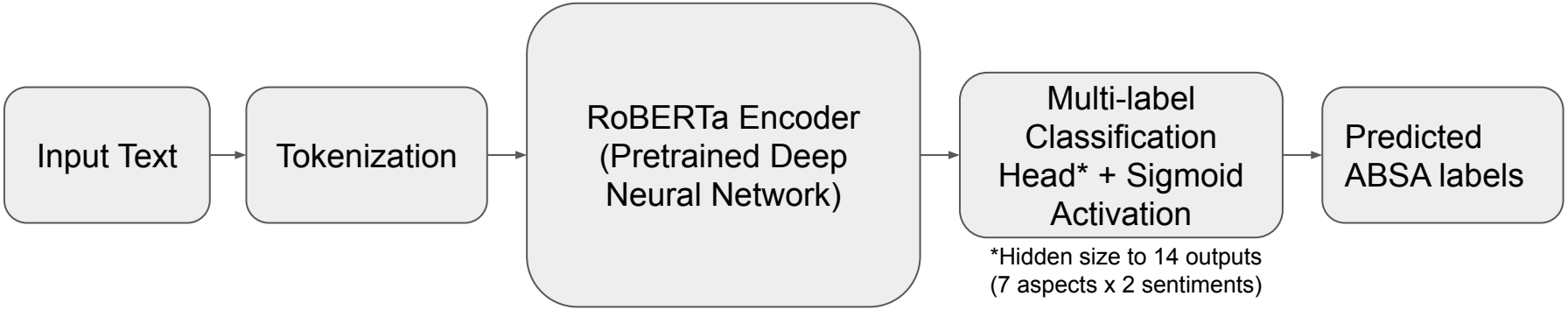
Model 1: Classical ML ABSA (baseline)

- Term Frequency - Inverse Document Frequency (TF-IDF) (sklearn)
- 90/10 train-test split (weakly supervised data)
- Aspect-sentiment extraction via Logistic Regression (sklearn)
  - MultiOutputClassifier trains separate binary classifiers for each of the aspect-sentiment labels, enabling LogisticRegressionCV to handle multi-label classification
  - Weighted classes to handle skew in ratings (see EDA)

```
clf_3 = MultiOutputClassifier(
    LogisticRegressionCV(
        cv=3,                          # Testing at 3 for speed
        scoring='f1_weighted',         # Optimize model
        class_weight={0: 1, 1: 2},     # 2x penalty for neg class
        random_state=2,                # Random seed
        n_jobs=1,                      # One CPU per classifier
        max_iter=500,                  # Max iterations
    )
```

Model 2: Neural transformer-encoder ABSA (primary)

- Tokenization of review text, specifically for RoBERTa (transformers, HuggingFace)
- RoBERTa neural transformer-encoder pretrained on 160GB of text, known for understanding language deeply



Model 2: Neural transformer-encoder ABSA (primary)

- Tokenization of review text, specifically for RoBERTa (transformers, HuggingFace)
- RoBERTa neural transformer-encoder trained on data and used to predict aspect-sentiment pairs

```
tokenizer = RobertaTokenizer.from_pretrained("roberta-base")

def tokenize_function(examples):
    """Tokenize text and attach sentiment labels to each row."""
    tokenized_dataset = tokenizer(
        examples['text'],    # Review text
        padding='longest',   # Pad reviews to longest in batch
        max_length=128,      # Max token length (98% of reviews were shorter)
        truncation=True      # Cut reviews longer than max_length
    )
    labels = []
    for i in range(len(examples['text'])):
        label_row = [float(examples[col][i]) for col in aspect_cols]
        labels.append(label_row)
    tokenized_dataset['labels'] = labels
    return tokenized_dataset
```

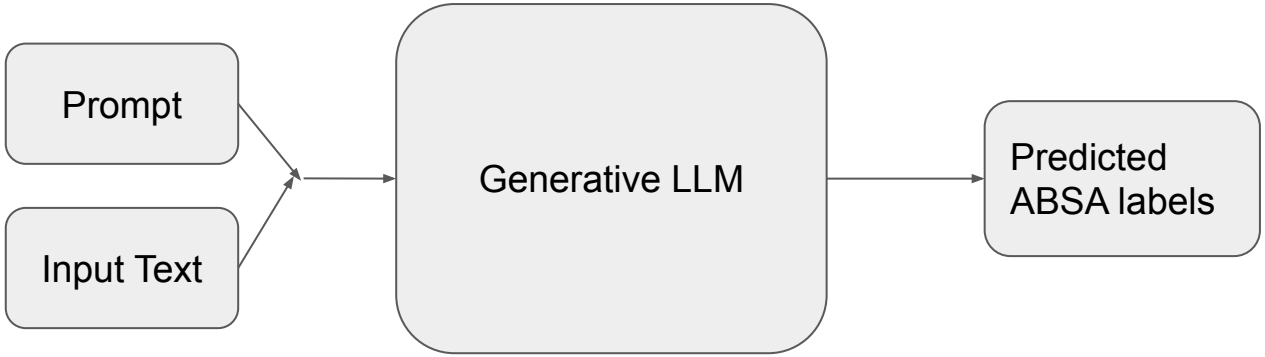
Model 2: Neural transformer-encoder ABSA (primary)

- Tokenization of review text, specifically for RoBERTa (transformers, HuggingFace)
- 90/10 train-test split
- RoBERTa neural transformer-encoder trained on data and used to predict aspect-sentiment pairs

```
SAMPLE_SIZE   = 100_000  # Sample size
TEST_SIZE     = 0.1      # 90/10 split
EPOCHS        = 3        # Dimishing returns past 3 (possible overfittting)
TRAIN_BATCH   = 64       # Reviews processed before updating weights
EVAL_BATCH    = 128      # Reviews evaluated at once
LEARNING_RATE = 2e-5     # Standard learning rate for RoBERTa
SEED          = 2        # Random seed
```

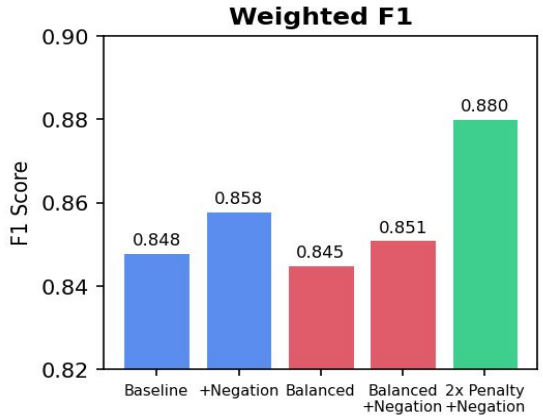
Model 3: Generative LLM (advanced/exploratory)

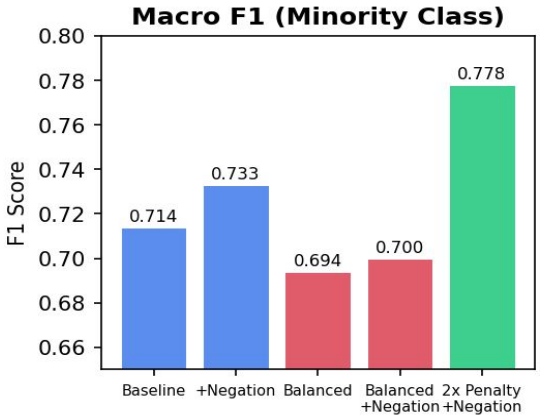
- No text cleaning or processing except for English-only
- Prompt generative LLM (e.g., GPT-5-mini) to “read” reviews and return aspect-sentiment pairs in a defined JSON format
- Use OpenAI’s API
- Zero-shot (prompt only) vs. Few-shot (prompt + examples)
- In Progress

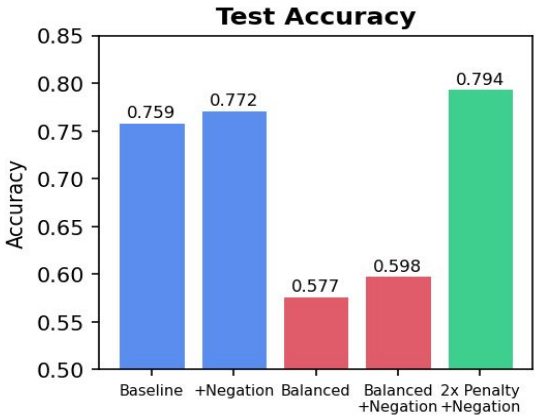


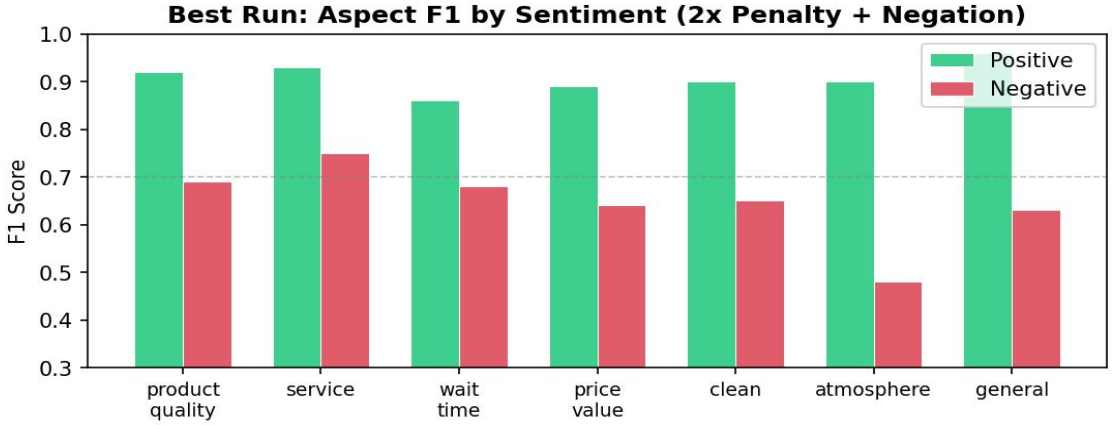
Results & Observations

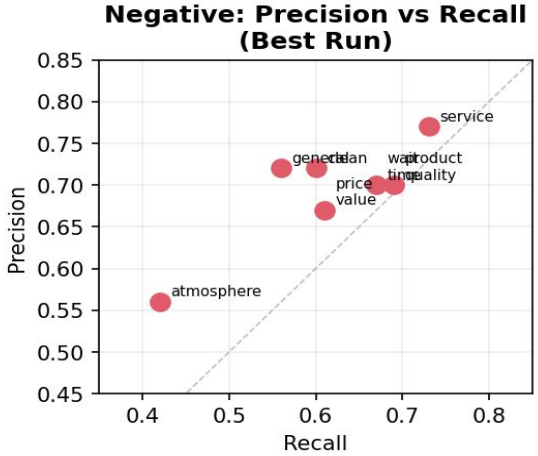
TF-IDF Logistic Regression — ABSA Results (22M Reviews)





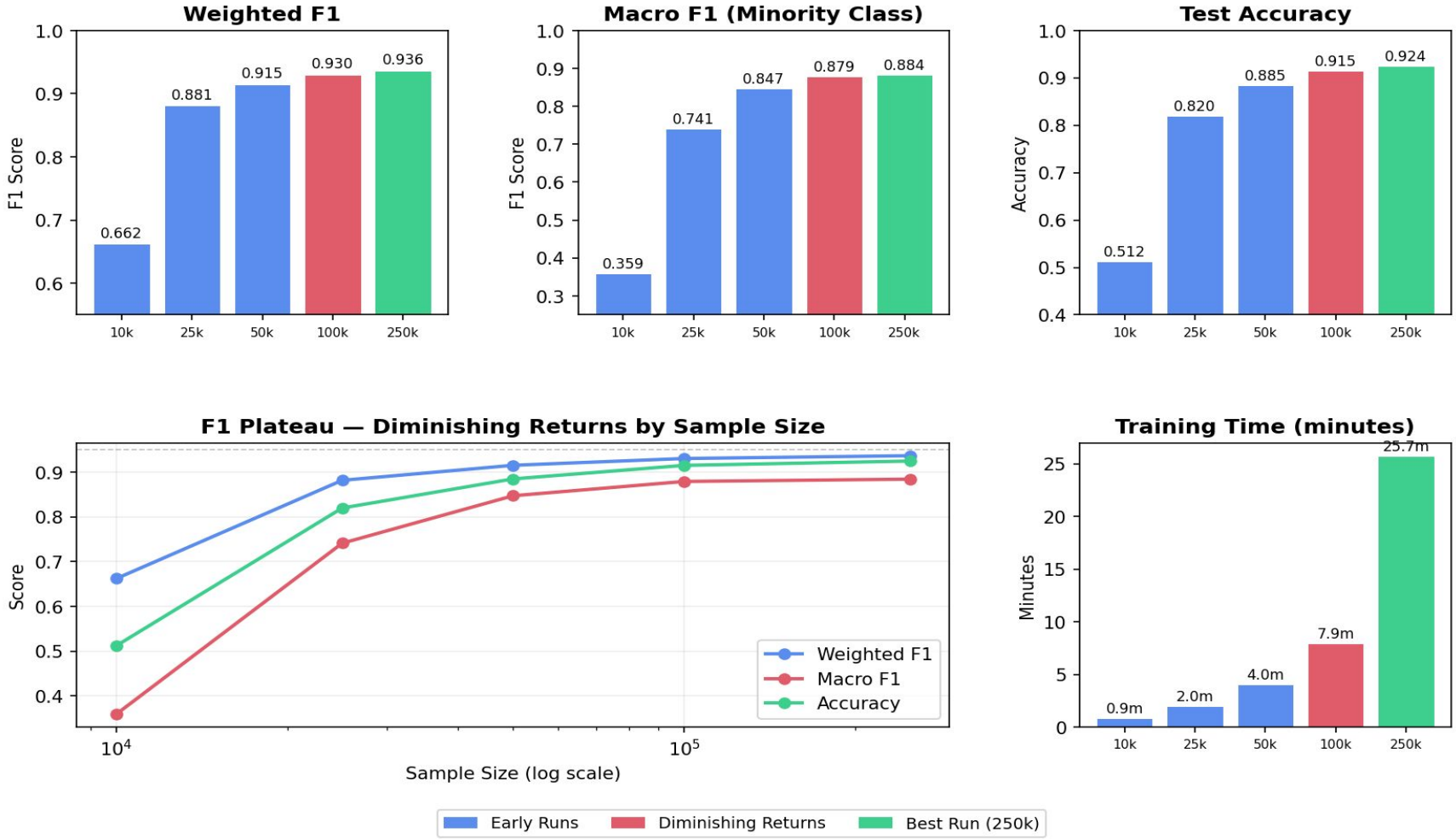






Baseline / Variants | Balanced (overcompensated) | Best: 2x Penalty + Negation

RoBERTa — ABSA Scaling Results (Sample Size vs Performance)



Progress Against Goals

- Goal 1: Clean data & process text ✅
- Goal 2: Select aspects ✅
- Goal 3: Apply weak supervision ✅
- Goal 4: Manually label subset of data for validation ❌
  - Gap: Need to finish labeling subset & run models
- Goal 5: Evaluate three different model types
  - Classical ML (baseline) ✅
  - Neural transformer-encoder (primary) ✅
  - Generative LLM (advanced/exploratory) ❌
    - Gap: Need to complete model pipeline & evaluate performance
- Goal 6: Business Owner Dashboard ❌
  - Gap: Use best model to extract top 3 successes and top 3 areas of improvement for a business

Section 2: Code Rundown

References

1. **Hu, M., & Liu, B. (2004).** *Mining and summarizing customer reviews.* Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD-04). Retrieved from https://www.cs.uic.edu/~liub/publications/kdd04-revSummary.pdf
2. **García-Pablos, A., Cuadros, M., & Rigau, G. (2017).** *W2VLDA: Almost unsupervised system for aspect-based sentiment analysis.* arXiv preprint arXiv:1705.07687. Retrieved from https://arxiv.org/abs/1705.07687
3. **Zhu, L., et al. (2022).** *Deep learning for aspect-based sentiment analysis: A review. Frontiers in Artificial Intelligence and Applications / PeerJ Computer Science* (survey article). Retrieved from https://pmc.ncbi.nlm.nih.gov/articles/PMC9454971
4. **Simmering, P. F., & Huoviala, P. (2023).** *Large language models for aspect-based sentiment analysis.* arXiv preprint arXiv:2310.18025. https://doi.org/10.48550/arXiv.2310.18025
5. **Kontonatsios, G., Clive, J., Harrison, G., Metcalfe, T., Sliwiak, P., Tahir, H., & Ghose, A. (2023).** *FABSA: An aspect-based sentiment analysis dataset of user reviews. Neurocomputing*, 562, 126867. https://www.sciencedirect.com/science/article/pii/S0925231223009906

As requested by the data source, the following papers are cited below:

UCTopic: Unsupervised Contrastive Learning for Phrase Representations and Topic Mining
Jiacheng Li, Jingbo Shang, Julian McAuley
Annual Meeting of the Association for Computational Linguistics (ACL), 2022
pdf

Personalized Showcases: Generating Multi-Modal Explanations for Recommendations
An Yan, Zhankui He, Jiacheng Li, Tianyang Zhang, Julian Mcauley
The 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2023
pdf